What kind of Natural Language Inference are NLP systems learning: Is this enough?

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Abstract: In this paper, we look at Natural Language Inference, arguing that the notion of inference the current NLP systems are learning is much narrower compared to the range of inference patterns found in human reasoning. We take a look at the history and the nature of creating datasets for NLI. We discuss the datasets that are mainly used today for the relevant tasks and show why those are not enough to generalize to other reasoning tasks, e.g. logical and legal reasoning, or reasoning in dialogue settings. We then proceed to propose ways in which this can be remedied, effectively producing more realistic datasets for NLI. Lastly, we argue that the NLP community could have been too hasty to altogether dismiss symbolic approaches in the study of NLI, given that these might still be relevant for more fine-grained cases of reasoning. As such, we argue for a more pluralistic take on tackling NLI, favoring hybrid rather than non-hybrid approaches.

1 INTRODUCTION

Reasoning is part of our every day routine: we hear Natural Language (NL) sentences, we participate in dialogues, we read books or legal documents. Successfully understanding, participating or communicating with others in these situations presupposes some form of reasoning: about individual sentences, whole paragraphs of legal documents, small or bigger pieces of dialogue and so on. The human reasoning performed in these different situations cannot be explained by a single rigid system of reasoning, plainly because reasoning is performed in different ways in each one of them. Consider the following example:

(1) Three representatives are needed.

If a human reasoner with expert knowledge was to interpret the above utterance in a legal context, s/he would most probably judge that a situation where more than three references are provided could be compatible with the semantics of the utterance. To the contrary, if the same reasoner was to interpret the above as part of a casual, everyday conversation, then *three* would most likely be interpreted as *exactly three*, making the same situation incompatible with the utterance. To give another example, consider the following dialogue interaction between participants A and B:

(2) A dialogue example

A. Mont Blanc is higher than

B. Mt. Ararat?

A. Yes.

B. No, this is not correct. It is the other way around.

A. Are you...

B. Sure? Yes, I am.

A. Ok, then.

The listener of this particular piece of dialogue will have to reason based on utterances that are split between two participants, thus having to dynamically keep track of them. Furthermore, the listener must be able, on the one hand, to compute global inferences, i.e. inferences that are based on statements/facts that are shared (agreed upon) by the dialogue participants and local inferences on the other, i.e. inferences that are based on facts that are not shared by all dialogue participants. Generalizing, we could say that the human ability to reason with Natural Language (NL), i.e. Natural Language Inference (NLI), cannot be seen as a single, coherent system of reasoning, but rather as a collection of reasoning tools, a toolbox to perform diverse reasoning tasks.

Even though current work in NLP can support a diversity of NLI scenarios, there is still a long way to go

to support the whole range of diversity found in NLI in general. Despite the usefulness of NLI, and huge steps made in the recent years, an important drawback remains in this line of work: NLI systems are evaluated against datasets which represent only a fraction of human reasoning possibilities. Furthermore, these different datasets seem to have arisen from the need to test specific theoretical architectures, for example, logical approaches in the case of the FraCaS test suite (Cooper et al., 1996), Deep Learning (DL) architectures in the case of Stanford Natura Language Inference Dataset (SNLI) (Bowman et al., 2015). What happens in practice is that any NLI system performs very poorly on any dataset which was not specifically intended to test it. As such, the different systems designed to tackle NLI are not only incomplete, but not even comparable.

This paper investigates the need of creating more realistic NLI datasets and argues for hybrid approaches to NLI that maintain a connection to symbolic NLP — contrary to current research trends. The structure of the paper is as follows: in section 2, the most prominent datasets used for NLI are presented and their respective advantages and/or weaknesses are discussed. In section 3, we ask the question of what kind of NLI systems trained on the existing datasets are learning and whether this is enough. Lastly, in section 4, we propose ways to create a more diverse and realistic collection of datasets, while we furthermore argue for the use of hybrid systems that retain connections with the symbolic world. More specifically, we argue that symbolic systems might still be relevant for more fine-grained NLI cases, e.g. logical or legal reasoning, making them useful as part of hybrid or controlled-domain systems for NLI.

2 NLI DATASETS

In this section, we go through the most prominent NLI datasets that have been used in NLP throughout the years. After this is done, we also briefly mention some datasets that, even though not NLI datasets *per se*, are quite useful and have been used for the study of NLI.

2.1 The FraCaS test suite

The FraCaS[1] test suite is an NLI data set consisting of 346 inference problems. Each problem contains one or more premises followed by one yes/no-question. There is a three way classification: YES, NO or UNK (unknown, see (3) for an example from FraCaS). The FraCaS test suite was later on turned into machine-readable format by Bill McCartney[2].

Extensions of FraCaS include: a) MultiFraCaS, in effect a multilingual FraCaS[3], and b) JSem, the Japanese counterpart to FraCaS, which expands the original FraCaS in a number of ways.[4]

Even though the FraCaS test suite contains a rather small number of examples (346), it covers a lot of NLI cases and is, at least to some extent, multilingual. On the downside, the suite includes mostly logical inferences. Furthermore, the size of the dataset is such that it cannot be used to train the Machine Learning nor Deep Learning (ML, DL) models.

(3) An UNK example from the FraCaS test suite.

P1 A Scandinavian won the Nobel Prize.

P2 Every Swede is Scandinavian.

H. Did a Swede win the Nobel prize?

H. A Swede won the Nobel prize.

Label UNK [FraCaS 065]

2.2 Recognizing Textual Entailment

The Recognizing Textual Entailment (RTE) challenges first appeared in 2004 as a means to test textual entailment, i.e. relations between a premise text and a hypothesis text (4):

(4) An entailment example from RTE1.

P. Budapest again became the focus of national political drama in the late 1980s, when Hungary led the reform movement in eastern Europe that broke the communist monopoly on political power and ushered in the possibility of multiparty politics.

H. In the late 1980s Budapest became the center of the reform movement.

Label Entailment [RTE702]

In contrast to the FraCaS test suite, the RTE challenges use naturally occurring data as premises. The hypothesis text is then constructed based on this premise text. There is either a binary or a tripartite classification of entailment — depending on the version of RTE. The first two RTE challenges follow the former scheme and make a binary classification of entailment

[1] ftp://ftp.cogsci.ed.ac.uk/pub/FRACAS/del16.ps.gz

[2] www-nlp.stanford.edu/ wcmac/downloads/fracas.xml.

[3] https://github.com/GU-CLASP/multifracas.

[4] More info on the suite and its innovations compared to the original FraCaS can be found here: http://researchmap.jp/community-inf/JSeM/?lang=english.

(entailed or not entailed). Tripartite classification (entailment, negation of the hypothesis entailment or no entailment) is added in the later datasets, retaining two way classification versions as well. Seven RTE challenges have been created altogether.

The main advantages of the RTE challenges is their use of examples from natural text and the inclusion of cases that require presupposed information. Another important characteristic is the inclusion of non-logical presuppositional inferences.

However, even though the RTE datasets have been notoriously difficult to tackle for NLI systems (especially the three-way entailment tasks), most of the examples do not involve any complex semantic inference. Rather, they are difficult to handle due to their use of the full range of natural syntax, and their dependence on world knowledge.

Indeed, the very *definition* of inference assumed in a number of the examples is problematic. As Zaenen et al. (2005) have pointed out, RTE platforms suffer from cases of inference that should not be categorised as such. For these cases, a vast amount of world knowledge needs to be taken into consideration (that most importantly not every linguistic agent has). The problem is that there is no clear annotation in the data that distinguishes the different kinds of inference. Furthermore, it is not clear whether the existence of background/hidden premises will be used by some speakers in order to classify a case as entailment, assuming that the presupposed information is plausible, or to the contrary render the case as non-entailment, assuming that the hidden premise is not plausible. Bernardy and Chatzikyriakidis (2018) show that validating RTE examples by asking subjects to provide justifications for their answers shows exactly that: some people may use a hidden assumption to justify an entailment, while some other subjects may use the same hidden premise to the contrary, i.e. to justify a non-entailment. Bernardy and Chatzikyriakidis (2018) asked expert linguists or logicians to validate a set of 130 examples taken from the RTE challenges that are marked as "YES". The subjects are asked to judge whether the conclusion follows or not from the premise, noting that in case extra assumptions need to be made to justify the answer, they should provide them. The results show that about half of the YES examples receive either a YES, IF..., a NO, BECAUSE..., or a straight NO answer.[5] Lastly, similarly to the FraCaS, the RTE datasets are still small (less than 1000 pairs for both the development and the test set for all challenges) with regards to datasets intended to train Deep Learning systems.

[5]https://github.com/GU-CLASP/PreciseTextualEntailment/blob/master/PilotEmail.txt.

2.3 SNLI, MultiNLI and XNLI

SNLI (Bowman et al., 2015) and MultiNLI (Williams et al., 2017) are two of the standard datasets used today to train and test Deep-Learning-based NLI systems. Both systems have been created using crowdsourcing techniques (Amazon Mechanical Turk). The process used to create SNLI is as follows: subjects are given a caption of a picture and then are asked to provide: a) an alternate true caption, b) an alternate possibly true caption, and c) an alternate false caption (figure 1). The dataset constructed out of this process contains 570k inference pairs, making SNLI two orders of magnitude bigger than datasets like FraCaS or RTE. MultiNLI was modeled on SNLI but uses data from a variety of genres. More specifically, ten different genres are represented from both written and spoken English. The dataset consists of 433k sentence pairs. Lastly, XNLI is a multilingual extension of MultiNLI (Conneau et al., 2018). It involves 5k test- and 2,5k dev-set examples from the MultiNLI translated into 14 languages. The size of SNLI and MultiNLI is suitable for training DL models, making them in this respect a very useful resource. Another defining characteristic is that reasoning in SNLI and MultiNLI is tied to specific situations (given by the picture captions).

While situational reasoning can be useful, it can also be a drawback of these datasets. An issue is that much reasoning involving quantifiers is not situational. What would be for example the image described by a caption "all men are human"? Similarly to earlier platforms, SNLI and MultiNLI seem to capture only a fraction of the range of phenomena associated with NLI. Even though MultiNLI claims to remedy this issue by introducing data from different genres, and indeed it definitely constitutes an improvement over SNLI, the definition of inference involved in both SNLI and MultiNLI is the same and quite narrow. For example, neither stricter/logical (or in general expert domain) reasoning of the sort found in the FraCaS test suite, nor the type of inference using world knowledge found in the RTE challenge, is found in any of SNLI or MultiNLI. Furthermore, the dialogue examples in MultiNLI involve clean cut cases of dialogue where the problematic aspects of it do not show up (e.g. split utterances, disfluencies etc.) and furthermore global vs local inferences are not checked. What appears to be further problematic in relation to SNLI is the containment of annotation artifacts. Specifically, Gururangan et al. (2018) show that both SNLI and MultiNLI contain annotation artifacts that help NN models in the classification task. For example, entailed hypotheses tend to contain generic words like *animal*, *instrument*, while contradicting hypotheses tend to involve negative quantifiers like *no*, *nobody* etc.

We will show you the caption for a photo. We will not show you the photo. Using only the caption and what you know about the world:

- Write one alternate caption that is **definitely** a **true** description of the photo. *Example: For the caption "Two dogs are running through a field." you could write "There are animals outdoors."*
- Write one alternate caption that **might be a true** description of the photo. *Example: For the caption "Two dogs are running through a field." you could write "Some puppies are running to catch a stick."*
- Write one alternate caption that is **definitely** a **false** description of the photo. *Example: For the caption "Two dogs are running through a field." you could write "The pets are sitting on a couch." This is different from the* maybe correct *category because it's impossible for the dogs to be both running and sitting.*

Figure 1: Instructions for the Mechanical Turk Data Collection.

2.4 SICK dataset

The Sentences Involving Compositional Knowledge (Marelli et al., 2014) is a dataset created, at least at its inception, to test compositional distributional semantics (DS) models. The dataset contains 9,840 examples of inference patterns (e.g. negation, conjunction, disjunction, apposition, relative clauses, etc.). However, it focuses on distributional semantic approaches. Therefore, it normalises several cases that DS is not expected to account for. The dataset is constructed by taking pairs of sentences from a random subset of the 8K ImageFlickr data set (Young et al., 2014) and the SemEval 2012 STS MSRVideo Description dataset.

2.5 Some other datasets related to NLI

There exist a number of NLI related datasets that have been used for NLI, which have received less attention from the community, so far. We mention some briefly here:

1. The QQP (Quora Question Pairs) dataset (Chen et al.) is an NLI dataset that contains pairs of questions from the Quora database and tries to classify them as semantically equivalent or not.
2. The PPDB (Paraphrase Database) relation extraction dataset (Ganitkevitch et al., 2015) is primarily a dataset on paraphrase. However, it is further annotated for entailment (unidirectional, bidirectional etc.), making it useful for entailment tasks as well.

Other datasets that are relevant for NLI, but are not NLI datasets *per se* include datasets on textual similarity like the Semantic Textual Similarity Benchmark (STS-B) (Cer et al., 2017), paraphrase datasets, the Microsoft Research Paraphrase Corpus (MRPC)[6], as well as answer sentence selection datasets (selQA) (Jurczyk et al., 2016). Detailing these datasets cannot be done here for lack of space: the interested reader is directed to the relevant papers for more information.

3 CAPABILITIES OF NLI SYSTEMS

SNLI and/or MultiNLI have been used as the dataset par excellence to train the latest state-of-the-art NLI models. This is not an accident, given that all the latest systems involve Neural Network (NN) architectures that require training sets sizes that that datasets like the FraCaS or RTE do not offer. The first system to be tested against SNLI achieved an accuracy of 0.8 using a vanilla Long-Short Term Memory Recurrent Neural Network (LSTM RNN) (Bowman et al., 2015). A number of other variations of LSTMs and bi-LSTMs improved the performance. For example, Wang et al. (2017) use a bilateral multi-perspective matching (BiMPM) model and achieve an accuracy of 0.888 on the SNLI dataset. This system uses a bi-LSTM to encode the inference pairs (P and Q), the pairs are further matched in both directions and then another bi-LSTM aggregates the results of this matching into a single vector that is used to make the final scoring decision. Chen et al. (2017a) achieve an accuracy of 0.891 on SNLI by enriching existing state-of-the art NLI models with external knowledge. External knowledge has been shown to work for earlier models based on logic or traditional ML techniques, but had never been used before for DL architectures. Tay et al. (2017) use a compare-aggregate architecture, where alignment features are propagated to higher layers and can thus be used. It provides an accuracy of 0.893 on the SNLI. The system presented by Kim et al. (2018), which is at the time of writing the state of the art on SNLI, uses a densely-connected recurrent network, in effect the RNN analogue of Densenet (Huang and Liu, 2017). Crucially, the recurrent features are retained all the way to the uppermost levels and a concatenation (rather than summation) operator works along the attention mechanism to preserve co-attentive information more efficiently. The system reports an accuracy of 0.901 on SNLI. Recently, Google's BERT system has been proposed, providing state-of-the-art results for 11 NLP benchmarks, among them the state-of-the-

[6] https://www.microsoft.com/en-us/download/details.aspx?id=52398.

Train	Dev	Test	Test Accuracy	Delta	Model
SNLI	**SNLI**	**SNLI**	**86.1**		600D BiLSTM-max
SNLI	**SNLI**	**SNLI**	**86.6**		600D HBMP Talman et al. (2018)
SNLI	**SNLI**	**SNLI**	**88.0**		600D ESIM Chen et al. (2017b)
SNLI	**SNLI**	**SNLI**	**88.6**		300D KIM Kim et al. (2018)
SNLI	SNLI	MultiNLI-m	55.7*	-30.4	600D BiLSTM-max
SNLI	SNLI	MultiNLI-m	56.3*	-30.3	600D HBMP
SNLI	SNLI	MultiNLI-m	59.2*	-28.8	600D ESIM
SNLI	SNLI	MultiNLI-m	61.7*	-26.9	300D KIM
SNLI	SNLI	SICK	54.5	-31.6	600D BiLSTM-max
SNLI	SNLI	SICK	53.1	-33.5	600D HBMP
SNLI	SNLI	SICK	54.3	-33.7	600D ESIM
SNLI	SNLI	SICK	55.8	-32.8	300D KIM
MultiNLI	**MultiNLI-m**	**MultiNLI-m**	**73.1***		**600D BiLSTM-max**
MultiNLI	**MultiNLI-m**	**MultiNLI-m**	**73.2***		**600D HBMP**
MultiNLI	**MultiNLI-m**	**MultiNLI-m**	**76.8***		**600D ESIM**
MultiNLI	**MultiNLI-m**	**MultiNLI-m**	**77.3***		**300D KIM**
MultiNLI	MultiNLI-m	SNLI	63.8	-9.3	600D BiLSTM-max
MultiNLI	MultiNLI-m	SNLI	65.3	-7.9	600D HBMP
MultiNLI	MultiNLI-m	SNLI	66.4	-10.4	600D ESIM
MultiNLI	MultiNLI-m	SNLI	68.5	-8.8	300D KIM
MultiNLI	MultiNLI-m	SICK	54.1	-19.0	600D BiLSTM-max
MultiNLI	MultiNLI-m	SICK	54.1	-19.1	600D HBMP
MultiNLI	MultiNLI-m	SICK	47.9	-28.9	600D ESIM
MultiNLI	MultiNLI-m	SICK	50.9	-26.4	300D KIM
SNLI+MultiNLI	**SNLI**	**SNLI**	**86.1**		**600D BiLSTM-max**
SNLI+MultiNLI	**SNLI**	**SNLI**	**86.1**		**600D HBMP**
SNLI+MultiNLI	**SNLI**	**SNLI**	**87.5**		**600D ESIM**
SNLI+MultiNLI	**SNLI**	**SNLI**	**86.2**		**300D KIM**
SNLI+MultiNLI	SNLI	SICK	54.5	-31.6	600D BiLSTM-max
SNLI+MultiNLI	SNLI	SICK	55.0	-31.1	600D HBMP
SNLI+MultiNLI	SNLI	SICK	54.5	-33.0	600D ESIM
SNLI+MultiNLI	SNLI	SICK	54.6	-31.6	300D KIM

Figure 2: Test accuracies (%). For results highlighted in bold the training data include examples from the same corpus as the test data. For the other cases, the training and test data involve separate corpora. Delta stands for the difference between the test accuracy and the baseline accuracy for the same training set. Results marked with * are for the development set, as no annotated test set is openly available. BiLSTM-max is Bidirectional LSTM with max pooling. HBMP is a Hierachical BiLSTM with max pooling.

art result for MultiNLI at 0.867 accuracy (Devlin et al., 2018).[7]

Within this context, and the impressive performance of NN systems w.r.t NLI tasks, two questions come to mind: a) what is the generalization ability of these systems, and b) what kind of inference are these systems learning. Both, we believe, are equally important. With regards to the first question, recent work on testing various state-of-the-art systems w.r.t their generalization ability has shown that it is rather limited. Glockner et al. (2018) have shown that NLI systems have limited generalization ability outside the datasets that they are trained and tested on. More specifically, they show that NLI systems break easily when, instead of being tested on the original SNLI test set, they are tested on a test set which contain sentences that differ by at most one word from sentences in the training set. A significant drop in accuracy, e.g. between 22 and 33 points when trained on SNLI and tested on the new dataset, is reported for three out of four state-of-the-art systems tested. The system less prone to breaking is Kim et al. (2018) (5 points drop when trained on SNLI and tested on the new dataset), which utilizes external knowledge taken from WordNet (Miller, 1995). Talman and Chatzikyriakidis (2018) train and test six state-of-the-art NN models using train and test sets drawn from a different corpus. For example, the train set is drawn from the SNLI but the test from the MultiNLI, vice versa and other similar combinations. The results shows an average drop of 24.9 points in accuracy for all systems, including the system by Kim et al. (2018). Results as reported by Talman and Chatzikyriakidis (2018), are shown in figure (2). The second question is directly relevant to the first one. What have these state-of-the-art models learned? It is obvious that they have learned something. However, whatever this is, does not seem to be generalizing well outside one specific dataset. One plausible explana-

[7]There are no results reported for SNLI.

tion is that the system has learned the very specific patterns of reasoning of specific datasets and not a generalized notion of inference *per se*. Will, then, a system trained on SNLI or MultiNLI be able to deal with more specialized cases of reasoning, like legal reasoning or any other kind of expert reasoning or reasoning in dialogue settings? The answer is most probably negative. As a hint to why this is so, consider the following example from SNLI:

(5) An example from SNLI

> **Premise:** A man selling donuts to a customer during a world exhibition event held in the city of Angeles.
>
> **Hypothesis:** A woman drinks her coffee in a small cafe.
>
> **Label:** Contradiction

The above example is labeled as a contradiction. But, there are a number of non-trivial steps to make in order to get to that conclusion: a) one has to assume that the two situations described are basically the same situation and in this sense try to see whether one description contradicts the other, b) the indefinite article in the premise has to be identified with the indefinite article in the hypothesis, thus *man* contradicting *woman* (a person cannot be a man and a woman at the same time). Such loose reasoning will not hold in cases where more precise reasoning might be needed, e.g. in legal contexts, where indefinites are generally understood as existential quantification. Thus, training systems using datasets with such a definition of entailment will most probably not be able to cope with more other cases of reasoning. To an extent, evidence of this exists in Talman and Chatzikyriakidis (2018), where training on SNLI or MultiNLI and testing on the SICK dataset, the latter involving a more strict definition of inference than the previous two datasets, gives a drop in accuracy between 19 and 33.6 points. To put these results into context, we move from systems that can be thought of as being useful in dealing with NLI, given that their accuracy is between 0.73 and 0.89, depending on the model and dataset, to systems that are not useful anymore, given that their accuracies are getting much closer to chance (.48 to .55 accuracy). Given these results and, furthermore, the idiosyncrasies of dialogue data and reasoning with those, it is safe to assume that state-of-the-art systems trained on SNLI or MultiNLI will not be able to generalize over reasoning with pieces of dialogue. On a more general note, and abstracting away from the individual datasets, it seems that what these systems of inference are learning, is a tiny fraction of what counts as human reasoning. And even in these cases, i.e. where a tiny fraction of NLI is taken to be NLI, generalization outside the specific datasets does not seem to happen.

3.1 Symbolic Systems for NLI: A Lost Cause?

Symbolic systems for NLI have been criticized as a means to deal with NLI, and NLP tasks in general, on the basis of coverage, i.e. the fact that these systems tend to easily break down once they are moved to open domains. This, as a general criticism, is of course to a great extent legit. It is true that symbolic/logical systems can be very precise, but have very poor recall, i.e. they break easily in the presence of new data. On the other hand, NN models have offered a hopeful way out of these problems, producing impressive results in all areas of NLP and most specifically, as already mentioned, in dealing with NLI. At first sight, these systems do not seem to be suffering from the brittleness problem just described for symbolic/logical approaches. This is to some extent correct, but not all the way though. For example, in the case of NLI, recent studies we have mentioned in section 2 show that state-of-the-art NLI systems are rather brittle as well, but brittle in another sense: they fail to generalize outside individual datasets and are, furthermore, unable to capture certain NLI patterns, at all. This is a very different kind of brittleness, different to the one found with symbolic/logical approaches to NLI. For example, it would be highly unlikely that a symbolic system would break down by creating a variation of a test dataset in the sense of Glockner et al. (2018), a dataset that managed to break a number of state-of-the-art NN NLI systems. On the other hand, building a logical system that is able to parse, in a reasonable way, a huge dataset like SNLI or MultiNLI, and produce reasonable logical forms is far from trivial. But on the assumption that this is somehow achieved, such a system will not be prone to the type of breaking shown by Glockner et al. (2018). At the same time, going to cases where a stricter definition of inference is assumed, e.g. logical inference, it is not clear that NN models will be able to stand up to the task. The difficulty here is that NN models should be able to somehow closely approximate or even worse model logics of some sort. And even though such research exists in the literature, i.e. using NNs to learn logical inference, the results are not conclusive. For example, Bowman et al. (2014) claim that DL systems can learn logic. They train their system on a task that involves pairs of simple sentences that have a logical relation to each other (for example, one simple pair could be "all reptiles walk" and "all turtles move"). Two recursive neural networks compute representa-

tions for the sentences. Then, these representations are fed into a simple feed-forward network that predicts the logical relation between the sentences. Veldhoen and Zuidema (2018) show, however, that such a claim is rather strong. NNs seem to learn local approximations rather than global solutions, which would be required to learn logical reasoning. Similar results are reported by Evans et al. (2018). It is an open question on whether NN models can actually, at this stage, deal with more strict, fine-grained cases of inference. Another issue that makes this task even more difficult are the datasets that would train NN models for this stricter inference. Given our discussion so far, it is fairly obvious that constructing such an expert dataset will have to rely, at least to a large extent, on expert judgments.

One should also realize that the standard NN models are limited already at the syntactic level. Indeed, for the relative simple task of matching agreement, several authors have found less than ideal accuracy (Linzen et al., 2016; Bernardy and Lappin, 2018). Bernardy (2018) also found significant drops in accuracy when generalizing to more than two extra nesting levels, even in the simplest case of a language comprised solely of parentheses.

State-of-the-art NN NLI models find limitations in the semantic level too, as the work of Lake and Baroni (2017) shows. They even fail to generalize between similar data sets. Indeed, Talman and Chatzikyriakidis (2018) show this when the training set is changed between datasets especially tailored for NN NLI models. The most striking cases are those where the systems are trained on SNLI and tested on MultiNLI and vice versa, given that these two datasets involve the same definition of inference (only difference is that MultiNLI is multi genre[8]). This is a more complicated problem. Even if a logical model that can deal with large datasets involving logical reasoning is constructed, it is not the case that it will be able to move to another dataset, where a different definition for inference is assumed. Asking a logical system to perform loose inference, as it is done in SNLI and MultiNLI, is probably too much to ask from the logic and even though some tweaking can be made to recover the hidden assumptions by injecting lexical knowledge, for datasets like SNLI, this seems to be a hopeless task. So, how is NLI to be handled? One way to think about the problem is to revisit earlier approaches to NLI, i.e. logical or vanilla ML approaches, and understand what individual approaches do well and what they do not. Given that we are dealing with a phenomenon that looks more like a toolbox to perform diverse reasoning tasks, rather than a single coherent reasoning system, looking at hybrid systems for NLI might be the optimal way to tackle the problem more efficiently.

[8]Note that multi genredness cannot explain why the systems fail, at least not completely. If this was the case, we would expect that the systems would fail one way, i.e. when trained on SNLI (single genre) and tested on MultiNLI (multi genre). This is however not the case as seen from the examples in figure 2.

4 TOWARDS MORE REALISTIC NLI DATASETS

What is a realistic collection of datasets for NLI? Based on the hypothesis that NLI is a much more complex phenomenon than NLP practitioners usually take it to be, one has to strike a balance between the diversity of reasoning tools found in actual human reasoning with NL and strive to successfully encode those in datasets that can be later used for training NLI systems. To be more precise, we attempt a categorization of types of reasoning based on five characteristics, shown below:

(6) Types of Reasoning involved in NLI

Types of Reasoning	
Situational	Non Situational
Precise	Loose
Open Genre	Genre Specific
External Knowledge-based	Self-contained
Dialogue	Non-Dialogue

These different type of reasoning translate to different features of datasets. Similarly, different systems are better in some of these types of reasoning, and worse in others. For example, let us take a state-of-the-art system like Google's BERT. What kind of reasoning has this system learned? BERT is evaluated against a number of datasets. The ones that are clearly NLI datasets though are MultiNLI and RTE. BERT does great on MultiNLI (state-of-the-art results, .867) and less good on RTE (0.701). Based on this data, the picture for BERT, as well as similar NN models is as follows:[9]

[9]MultiNLI involves dialogue data, but reasoning in these cases is not contingent on the intricacies of dialogue, so at least with respect to reasoning, there is no dialogue aspect. Furthermore, NN systems seem to be tackling some world knowledge based reasoning, but this is not what they do very well, so we are reluctant to assume that they are good at this type of reasoning.

(7) State-of-the-art NN NLI models

NN NLI models	
Situational ✓	Non Situational
Precise	Loose ✓
Open text ✓	Controlled text ✓
External Knowledge-based	Self-contained ✓
Dialogue	Non Dialogue ✓

What are systems based on logic doing? For example, let us take the recent system of Bernardy and Chatzikyriakidis (2017). This is a logical system, which achieves an accuracy of .83 on approximately half of the FraCaS test suite. What FraCaS is capturing, and what this system, as well as and similar systems based on logic are capturing is shown below:

(8) State-of-the-art Logical models of NLI

State-of-the-art Logical Models	
Situational	Non Situational ✓
Precise ✓	Loose
Open text	Controlled text ✓
External Knowledge-based	Self-contained ✓
Dialogue	Non Dialogue ✓

The two different systems are complementary with respect to the first three points. However, the question is whether some of these are artifacts of the fact that these systems are evaluated on different datasets. This is correct to some extent. For example, SNLI only has situational reasoning, while the FraCaS non situational. But you could imagine extensions of both datasets to include the other options as well. This is actually the case with MultiNLI, the multi-genre extension of SNLI, where reasoning there is non situational. The second aspect, involving precise and loose reasoning, is more difficult. Datasets actually differ in their definition of NLI and whether it should be precise or not. SNLI follows the latter, FraCaS the former. The problem is that systems capturing one of the two, will most probably not be able to accommodate the other. The third aspect relates to inherent difficulties of logical systems and the fact that logical systems are very brittle, namely they fail on open text. NN models fare much better in this respect.[10] Both approaches have difficulties for knowledge based reasoning. None of the two can capture aspects of reasoning with dialogue in NLI. Actually, at the moment, no dataset capturing this sort of reasoning in dialogue exists, so any kind of comparison is not possible. This also brings up the need for the construction of such dataset(s).

[10] But can be claimed to be brittle in another sense, as we have argued in the previous section.

Having as a starting point this rough break down of NLI into individual aspects that can take two values, we argue that general-purpose datasets for NLI should involve (at least) all these options:

(9) Reasoning cases to be captured by an NLI dataset

Types of Reasoning for better NLI datasets	
Situational ✓	Non Situational ✓
Precise ✓	Loose ✓
Open Text ✓	Controlled Text ✓
External Knowledge-based ✓	Self-contained ✓
Dialogue ✓	Non Dialogue ✓

Conversely, when proposing a new dataset to test NLI systems, one should attempt to categorize it according to the above (or be even more precise).

In order to achieve the goal of a full-coverage dataset, we believe that one has to use a combination of techniques for data collection and validation, rather than a single method:

- Expert judgments
- Crowdsourcing using crowdsourcing platforms like MT or Crowdflower
- Crowdsourcing using Games with a Purpose (GWAPs)

The aspects situational/non situational and open text/controlled text are not very difficult to achieve. Datasets like SNLI involve situational reasoning, while MultiNLI, which can be seen as the multi genre extension of SNLI includes non situational reasoning. They can be both seen as Open text. It is not very hard to assume data collection combining the two, thus the end result involving both situational and non situational reasoning.[11] Crowdsourcing platforms like MT or Crowdflower might be enough in this case.

The criteria of precise/loose and external knowledge-based/self-contained are trickier. First of all, it seems that expert judgments will be necessary when designing datasets for more fine-grained/precise reasoning. For example, imagine that you want to train a system to deal with legal text, so that given a set of legal premises and a conclusion, it can decide whether the latter follows from the former or not. Obviously, the dataset used for this purpose has to be constructed by specialists that know how to reason within this specialized domain. This might be a costly task compared to the use of cheap labor via platforms like Mechanical Turk (MT), but if we are to get any real sense of the complexity and the domain specificity of NLI, such tasks need to be

[11] Actually, it is not very uncommon to train NLI systems on both SNLI and MultiNLI at the same time. This pretty much has the effect described.

performed and at least to some extent, experts have to be used. Furthermore, and connected to the external knowledge-based/self-contained aspect, even every day reasoning can get more fine-grained depending on how much time the agent is willing to spend in thinking about the inference patterns. For example, look at the following example from RTE3:

(10) RTE3, Problem Number 343

> **Premise:** November 9, 1989, the day the Berlin Wall fell and the world changed forever. Not even the most astute saw it coming. As Hungary's foreign minister in the late summer of 1989, Gyula Horn gave the order to let visiting East Germans use his country to do a 400-mile end run around the Berlin Wall, a move now seen as the beginning of the end for hard-line communism in Europe.
>
> **Hypothesis:** The Berlin Wall was torn down in 1989.
>
> **Label:** YES

The problem here is whether the reasoner thinks that *fell* can be coerced into implying *tear down*. Asking 3 expert linguists and one logician to label the example indeed brought the issue up: two of them labeled it as *Yes, If fell implies tear down* and one marked it as *No, because fell does not imply tear down*. Here is another example from Bernardy and Chatzikyriakidis (2018) that is quite representative of the situation:

(11)

> P: Philip Morris the US food and tobacco group that makes Marlboro, the world's best-selling cigarette, shrugged off strong anti-smoking sentiment in the US.
> H: Philip Morris owns the Marlboro brand.
> A1: Yes, if making involves owning the brand
> A2: Yes, if making something implies owning the brand
> A3: Yes, if a company making a product, owns the brand of the product
> A4: No, because making the product does not imply owning the brand

These examples show that it is very hard to know what kind of background hidden premises count as more or less safe to assume and which ones are not. Not only different people will have different opinions, but, also, the same people will have different opinions if you ask them to think more carefully about example pairs. One way to deal with this problem is to collect the same external knowledge premises and then count instances in which they have been used as supporting the inference, and instances where the same ones are used not supporting the inference. For example, in (11), the premise *making the product implies owning the brand* is used to justify an entailment by three annotators, whereas its negation, *making the product does not imply owning the brand*, is used by one annotator to justify a non-entailment. Having large scale expert annotation, where at least four expert annotators are involved, can give us a way to construct knowledge based NLI examples by counting the times implicit premises are used to justify an entailment, counting the times the same implicit premises are used to justify a non-entailment, and then checking whether the first number is more than half: if it is, then the example is included as an entailment case along with its backgrounded world knowledge premise. It is obvious that using MT, at least in the way it has been used so far for NLI data collection, will not provide us with this level of precision[12] So, large scale expert annotation would be necessary for more fine-grained or external knowledge-based NLI cases. The current NLI datasets do not reflect this more fine-grained aspect of reasoning.

The last aspect, dialogue/non dialogue, is not covered at all in NLI datasets. The datasets we have so far are constructed on the basis of complete sentences/pieces of text pairs. However, language is rarely that clean cut in everyday linguistic interaction. For example, in conversation, quite often, we do not speak in complete sentences. What one thinks as "complete sentences" usually emerge through a sequence of subsentential contributions. Each interlocutor potentially adds more structure to an already partial one during turn-taking:

(12) A dialogue example

> A. Mont Blanc is higher than
>
> B. Mt. Ararat?
>
> A. Yes.
>
> B. No, this is not correct. It is the other way around.
>
> A. Are you...
>
> B. Sure? Yes, I am.
>
> A. Ok, then.

Despite the fragmentary nature of dialogue, humans are able to perform reasoning tasks at each stage of the interaction and update these inferences if needed

[12] Or to put it somewhat more mildly: it will be difficult to set the task in a way that will capture the fine-grainedness level intended. Other online crowdsourcing tools, like semant-o-matic as designed by Dobnik and used for NLI by Chatzikyriakidis et al. (2017a)) seem to be better fit for such a task.

when more information comes in. As far as we know, there is no entailment dataset for dialogue data, and thus no dataset that will include reasoning with this type of data. Given that dialogue data is a core part of NL, this is something that NLP researchers need to start thinking about at some point. The question remains: how is this to be done? One way to do it, at least as a starting point, is to build such datasets via extracting dialogue pieces from corpora like the British National Corpus (BNC) or the newest dialogue datasets, most prominently bAbI (Bordes et al., 2016) and bAbI+ (Shalyminov et al., 2017), use the pieces as premises and then construct the hypothesis based on those. Given the nature of the task, issues like participants' individual beliefs will come into play. For example, here are a number of hypotheses constructed out of the previous artificial dialogue piece (12):

(13) A formed hypothesis against a fragment of the dialogue piece

Hypothesis: A and B believe that Mt. Ararat is higher than Mont Blanc

Label: Entailment

However, note that in case the dialogue piece we use as a premise is a fragment of the original one, as shown below, then the entailment does not hold:

(14) Formed hypothesis against the full dialogue piece

Premise:

A. Mont Blanc is higher than

B. Mt. Ararat?

A. Yes.

B. No, this is not correct. It is the other way around.

Hypothesis: A and B believe that Mt. Ararat is higher than Mont Blanc.

Label: Non-entailment

To give a real example, consider the following example constructed using a dialogue piece from bAbI+, an extension of the bAbI dataset, as premise. The latter contains goal-oriented dialogues in the domain of restaurant search, and the former expands a subpart of bAbI, everyday incremental dialogue phenomena (e.g. hesitations, restarts, and corrections):

Figure 3: Screenshot of an ongoing game with the target noun *fromage* (cheese). Several suggestions have been proposed by the user. These are listed on the right hand side.

(15) An NLI example based on a bAbI+ example

Premise:

sys hello what can I help you with today?

usr Id like to book a uhm yeah Id like to book a table in a expensive price range

sys Im on it. Any preference on a type of cuisine

usr with indian food no sorry with spanish food please

Hypothesis: The user wants to eat Spanish food.

Label: Entailment.

4.1 Using Serious Games to Complement NLI Data Collection

Serious Games or Games With a Purpose (hereafter GWAP) have been used successfully in collecting linguistic data. A prominent example is the GWAP *JeuxDeMots* (JDM, Lafourcade et al. (2015). JDM is a two-player GWAP, where participants *earn and collect* words. The main mechanism to achieve this, is the provision of lexical and semantic associations to terms that the system. proposes. The intended reader is directed to Lafourcade and Joubert (2008); Chatzikyriakidis et al. (2017b) for more information. JDM has grown up to include more than 1M terms and more than 230M lexical relations. The system is very fine-grained: it is based on 100 pre-defined relation types.

Using GWAPs might be a solution in order to deal, at least to some extent, with the problems outlined. For example, one could envision a GWAP where players, similarly to proposing lexical relations, are given a (premise,hypothesis) pair instead and are asked to provide all the hidden assumptions used in order for the pair to be labelled as an entailment. In the case of

dialogue data, one can construct a collaborative game where the players are asked to provide continuations for fragments of dialogue. Once a database of dialogue pieces is constructed, another part of the game might involve the task of providing inferences based on the pieces of dialogue just constructed. Of course, creating such a game is not straightforward, since in order to be successful, a number of parameters have to be taken into consideration. As Alain Joubert and Brun (2018) point out, GWAPs have to:

1. be attractive, fun and interesting.
2. be easy to understand with a simple set of instructions.
3. be addictive
4. allow filtering of players. This translates to making the players feel important when playing and guilty in case they are not playing for some time.

Serious games can also be used as a means to validate existing datasets. For example, one way to validate entailment pairs is to model the task-taking ideas from the construction of JDM and its "satellite games". Satellite games are a number of GWAPs peripheral to JDM that eventually, besides collecting data on their right, also feed information and help further develop JDM. For example, in LikeIt[13], the players are given a term and are asked to describe their sentiments towards it (positive, neutral, negative) and in AskIt[14], players are given two terms along with a question about a potential relation between the two and are asked to mark whether this relation exists, does not exist or maybe exists (YES, NO, POSSIBLY). What is important, is that there seems to be a lot of knowledge and experience in the field for how to build systems that will successfully produce good quality data for a given task. It remains to be shown in practice by creating a GWAP that will collect NLI data.

5 CONCLUSIONS

In this paper we investigated the type of inference state-of-the-art NLI systems are learning. Firstly, it has been argued that current datasets for NLI are far from reflecting the complexity of NLI and some clear examples exemplifying why this is so were presented. Then, we have claimed that a result of this situation is that state-of-the-art systems will not be able to generalize across different NLI cases, where a slightly different definition of inference is involved. This is partially borne out by the experimental results of Talman and Chatzikyriakidis (2018). Furthermore, and even worse, state-of-the-art NLI models do not seem to be able to generalize outside specific datasets, even when the same definition of NLI is assumed. We then presented some ideas of how one can build more realistic NLI datasets that can be used to better reflect NLI and potentially help in developing better NLI models. Lastly, we have discussed the use of symbolic/logical approaches to NLI and argued that the NLP community has been probably too hasty in dismissing them as a candidate solution for NLI.

[13] http://www.jeuxdemots.org/likeit.php

[14] http://www.jeuxdemots.org/askit.php?pass=1

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